

A Reproducible Walk-Through of RSA, Classical Factoring, and Shor’s Algorithm on IBM Heron r2 Hardware

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Abstract—This article gives an end-to-end explanation of the Rivest–Shamir–Adleman (RSA) cryptosystem [1], classical integer-factorisation attacks against RSA [2], [3], and the quantum order-finding method introduced by Shor [4], [5]. The discussion is paired with a small, notebook-backed demonstration for the standard teaching modulus $N = 15$: the same public modulus is factored by trial division, by Pollard’s ρ method, by a noiseless Qiskit Aer simulation, and by one execution on `ibm_kingston`, a 156-qubit IBM Heron r2 processor [6], [7]. The result is not a cryptographically meaningful break of RSA; it is a traceable demonstration of how the mathematics, circuit model, hardware noise, and post-processing fit together at small scale. The article distinguishes this toy experiment from resource estimates for RSA-2048, including the Gidney–Ekerä estimate of roughly eight hours using about 2×10^7 noisy physical qubits under specified surface-code assumptions [8] and a later Gidney preprint that lowers the qubit estimate below one million while increasing the runtime under similar assumptions [9]. It also connects the long-term risk to harvest-now, decrypt-later planning [10] and to the NIST post-quantum standards FIPS 203, FIPS 204, and FIPS 205 [11]–[13].

Index Terms—RSA, public-key cryptography, integer factorisation, Shor’s algorithm, quantum phase estimation, NISQ, post-quantum cryptography, IBM Heron r2, harvest-now-decrypt-later.

I. INTRODUCTION

For nearly five decades, public-key cryptography has supported digital identity, secure messaging, e-commerce, and government communications. The Diffie–Hellman key-exchange method [14] and RSA [1] showed that practical public-key systems could be built from problems believed to be hard for classical computers. In RSA, the relevant hardness assumption is integer factorisation: given a large composite modulus $N = pq$, recovering the prime factors p and q is not known to be feasible for classical machines when N is generated with modern key sizes.

Shor’s 1994 algorithm changed the theoretical landscape by giving a polynomial-time quantum algorithm for integer factorisation and discrete logarithms [4], [5]. That result does not imply that deployed RSA can be broken today. A cryptographically relevant attack would require a fault-tolerant quantum computer with enough logical

qubits and enough protected physical qubits to run a deep order-finding circuit reliably. Gidney and Ekerä estimated one concrete surface-code design for RSA-2048 at about eight hours of runtime using about 2×10^7 noisy physical qubits under their stated hardware assumptions [8]. A later Gidney preprint estimates less than one million noisy physical qubits, but with a runtime below one week rather than eight hours under similar modelling assumptions [9]. Current IBM systems include 156-qubit Heron r2 and r3 processors [7], which remain noisy intermediate-scale quantum (NISQ) devices in the sense described by Preskill [15].

This article has a limited goal. It does not introduce a new cryptanalytic attack, a new circuit compiler, or a new hardware result. Instead, it provides a citation-traceable explanation and small-scale experiment covering:

- (i) RSA key generation, encryption, and decryption;
- (ii) classical factoring by trial division, Pollard’s ρ method, and the General Number Field Sieve (GNFS) [2], [3];
- (iii) Shor’s reduction from factoring to quantum order finding [5];
- (iv) notebook-backed $N = 15$ executions in a noiseless simulator and on `ibm_kingston`, a 156Q Heron r2 backend announced by IBM in 2025 [6]; and
- (v) the migration context created by harvest-now, decrypt-later risk, NSM-10, CNSA 2.0, and the NIST PQC standards [10]–[13], [16], [17].

The deliberately small modulus keeps the arithmetic and circuit behaviour visible. It should not be read as evidence that present NISQ hardware can attack production RSA keys.

II. THE RSA CRYPTOSYSTEM

A. Algebraic Setting

RSA operates on the multiplicative group of integers modulo a composite $N = pq$ where p and q are distinct prime numbers. Its correctness rests on the following corollary of Euler’s theorem [1]: for any integer m coprime to N ,

$$m^{\varphi(N)} \equiv 1 \pmod{N}, \quad (1)$$

Algorithm 1 Textbook RSA, key generation and use

- 1: Pick distinct primes p, q
 - 2: $N \leftarrow pq$; $\varphi(N) \leftarrow (p-1)(q-1)$
 - 3: Pick e with $\gcd(e, \varphi(N)) = 1$
 - 4: $d \leftarrow e^{-1} \pmod{\varphi(N)}$
 - 5: **Public key:** (N, e) **Private key:** d

 - 6: **Encrypt** integer $m \in (0, N)$: $c \leftarrow m^e \bmod N$
 - 7: **Decrypt** integer c : $m \leftarrow c^d \bmod N$
-

where $\varphi(N) = (p-1)(q-1)$ is Euler’s totient. Choosing a public exponent e coprime to $\varphi(N)$ and a private exponent $d \equiv e^{-1} \pmod{\varphi(N)}$ yields the central RSA identity:

$$(m^e)^d \equiv m^{ed} \equiv m^{1+k\varphi(N)} \equiv m \pmod{N}. \quad (2)$$

B. Key Generation, Encryption, and Decryption

Algorithm 1 summarises textbook RSA key generation, encryption, and decryption. In this simplified presentation, encryption and decryption are each one modular exponentiation.

C. Canonical Pedagogical Instance: $N = 15$

For pedagogical and experimental uniformity, this paper fixes the parameters

$$p = 3, \quad q = 5, \quad N = 15, \quad \varphi(N) = 8, \quad e = 3, \quad d = 3, \quad (3)$$

and we use Alice’s secret integer $m = 8$, giving ciphertext

$$c = m^e \bmod N = 8^3 \bmod 15 = 512 \bmod 15 = 2. \quad (4)$$

At a modulus of four bits the security of the scheme is vacuous; the purpose of $N = 15$ is solely to provide an instance small enough to be factored classically in microseconds and used in a small quantum order-finding circuit on present-day NISQ hardware. We return to this point in Section VI.

D. Encryption as a Permutation

For fixed (N, e) with $\gcd(e, \varphi(N)) = 1$, the map $m \mapsto m^e \bmod N$ is a bijection modulo N . Decryption is its inverse permutation. Figure 1 visualises the non-zero residues used in the canonical instance. The figure makes plain that every displayed plaintext lands on a unique ciphertext and is recovered by the corresponding inverse permutation, showing that decryption reverses encryption for each valid residue.

E. The Cryptographic Bet

RSA security depends on the computational assumption that, given only N , the recovery of its prime factors p and q is computationally infeasible. Every other component of the scheme is mechanical: knowledge of (p, q) immediately yields $\varphi(N)$, hence d , hence m . The rest of the article compares how this assumption behaves in classical and quantum computational models.

III. CLASSICAL CRYPTANALYSIS

A. Trial Division

The naive baseline is trial division: enumerate candidate divisors $d = 2, 3, \dots, \lfloor \sqrt{N} \rfloor$ and test divisibility. Its time complexity is

$$T_{\text{TD}}(N) \in \mathcal{O}(\sqrt{N}) = \mathcal{O}(2^{n/2}), \quad (5)$$

where $n = \lceil \log_2 N \rceil$ is the bit-length of N . At $N = 15$ a single odd candidate suffices. At $n = 2048$, the exponential cost makes this approach irrelevant for real RSA keys.

B. Pollard’s ρ Algorithm

A markedly better classical algorithm is Pollard’s ρ method, which iterates the polynomial map

$$x_{i+1} = x_i^2 + c \pmod{N}, \quad c \neq 0, -2, \quad (6)$$

and uses Floyd’s cycle detection (the “tortoise and hare”) to find two indices $i \neq j$ with $x_i \equiv x_j \pmod{p}$. Whenever such a coincidence arises, $\gcd(|x_i - x_j|, N)$ is, with high probability, a non-trivial factor of N . By the birthday bound the expected runtime is [2], [18]

$$T_\rho(N) \in \mathcal{O}(N^{1/4}) = \mathcal{O}(2^{n/4}), \quad (7)$$

a quadratic speed-up over trial division but still exponential in n . For $n = 2048$ this is $\Theta(2^{512})$ operations: again, infeasible.

C. The General Number Field Sieve

The best classical algorithm presently known for factoring large semiprimes is the *General Number Field Sieve* (GNFS), whose runtime obeys the sub-exponential L -notation expression

$$L_N \left[\frac{1}{3}, \sqrt[3]{\frac{64}{9}} \right] = \exp \left(\left(\sqrt[3]{\frac{64}{9}} + o(1) \right) (\ln N)^{1/3} (\ln \ln N)^{2/3} \right). \quad (8)$$

GNFS is the algorithm that has set every modern factoring record: RSA-768 (232 decimal digits) was broken in 2010 after the equivalent of roughly 2000 CPU-years on a single 2.2 GHz Opteron core [19], and RSA-240 (795 bits) was reported in 2020 [20]. Larger RSA challenge moduli such as RSA-1024 and RSA-2048 have no public factorisations reported in the cited record literature [19], [20].

D. Empirical Demonstration at $N = 15$

Figure 2 reports the notebook-recorded wall-clock time for trial division and Pollard’s ρ on the canonical $N = 15$. Both finish in microseconds because the modulus is absurdly small; the figure is included only to show the relative behaviour of the two routines on the same toy input. Table I summarises selected public factoring records and the status of larger RSA moduli in the public record considered here.

The public record considered here contains no reported classical factorisation of RSA-1024, RSA-2048, or RSA-4096. The relevant theoretical contrast is that Shor’s

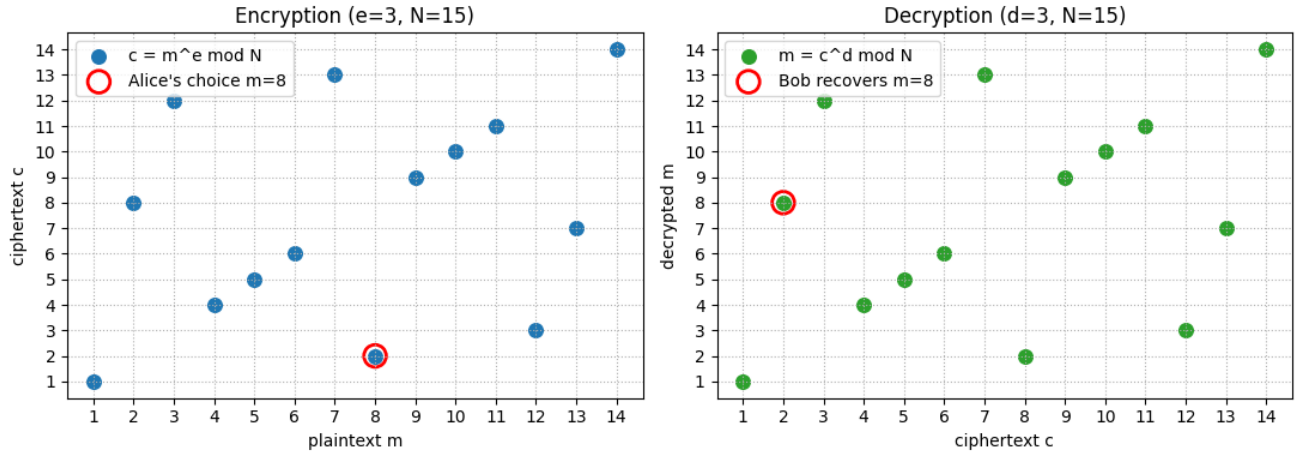


Fig. 1: RSA viewed as a bijection on $\{1, \dots, N-1\}$ for $N=15$, $e=3$. Left: encryption $c = m^e \bmod N$. Right: decryption $m = c^d \bmod N$ with $d=3$. The red marker identifies Alice's chosen plaintext $m=8$ and the ciphertext $c=2$.

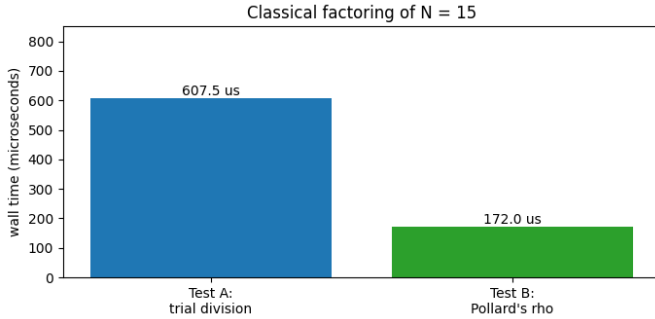


Fig. 2: Wall-clock time (microseconds) for the two classical factoring routines on $N=15$. The cost gap is invisible at this modulus but the asymptotic scaling diverges catastrophically as N grows.

TABLE I: Selected public factoring records and larger RSA moduli.

Modulus	Algorithm	Public status
RSA-512	NFS/GNFS	Factored in 1999 [21]
RSA-768	GNFS	Factored in 2010 [19]
RSA-240	GNFS	Factored in 2020 [20]
RSA-1024	GNFS	No public factorisation reported [22]
RSA-2048	GNFS	No public factorisation reported [8]
RSA-4096	GNFS	No public factorisation reported [8]

algorithm is polynomial-time in the input size, while the best known general-purpose classical factoring method is sub-exponential [3], [5].

IV. QUANTUM CRYPTANALYSIS: SHOR'S ALGORITHM

A. Reduction of Factoring to Order Finding

Shor reduces integer factorisation to the abelian *order-finding* problem: given N and a with $\gcd(a, N) = 1$, find the least integer $r \geq 1$ such that

$$a^r \equiv 1 \pmod{N}. \quad (9)$$

If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then with probability at least $1/2$ over random choice of a [5],

$$p = \gcd(a^{r/2} - 1, N), \quad q = \gcd(a^{r/2} + 1, N) \quad (10)$$

are non-trivial factors of N . This classical reduction is efficient. The hard step, intractable classically, is the determination of r .

B. Quantum Phase Estimation as Order-Finding

Shor's insight is that order finding reduces to *quantum phase estimation* (QPE) on the modular-multiplication unitary

$$U_a |y\rangle = |ay \bmod N\rangle. \quad (11)$$

The eigenvalues of U_a on the cyclic subgroup generated by a have the form $e^{2\pi i s/r}$ for $s \in \{0, 1, \dots, r-1\}$. The QPE subroutine, originally formulated by Kitaev [23] and analysed in detail in [24], prepares an equal superposition over the t -qubit counting register, applies the controlled- $U_a^{2^k}$ operators for $k = 0, 1, \dots, t-1$, applies an inverse Quantum Fourier Transform (QFT) [25], and measures. The measurement returns an integer m_ϕ approximating $2^t \cdot (s/r)$. *Continued fractions* [5], [26] then recover r from $m_\phi/2^t$ with probability bounded below by a constant.

C. Algorithmic Skeleton

Algorithm 2 captures the full procedure.

D. Time and Space Complexity

The textbook compilation of Shor's algorithm uses $\mathcal{O}(n)$ counting qubits and $\mathcal{O}(n)$ work qubits, where $n = \lceil \log_2 N \rceil$, and executes a number of elementary gates polynomial in n ; textbook presentations commonly describe the modular-exponentiation cost as $\mathcal{O}(n^3)$ arithmetic-scale operations [5], [24]. Beauregard subsequently demonstrated a semiclassical compilation requiring only $2n+3$ qubits [27]. Häner, Roetteler, and Svore refined the space requirement to $2n+2$ qubits; their Toffoli-based modular

TABLE II: Asymptotic factoring complexity: classical vs. quantum.

Algorithm	Model	Time complexity in $n = \log_2 N$
Trial division	Classical	$\mathcal{O}(2^{n/2})$
Pollard's ρ	Classical	$\mathcal{O}(2^{n/4})$
GNFS [3]	Classical	$\exp\left(\left(\sqrt[3]{\frac{64}{9}} + o(1)\right)n^{1/3}(\ln n)^{2/3}\right)$
Shor (textbook) [5]	Quantum	$\mathcal{O}(n^3)$
Shor (HRS) [28]	Quantum	depth $\mathcal{O}(n^3)$; gate count $\mathcal{O}(n^3 \log n)$

Algorithm 2 Shor's algorithm for integer factorisation [5]

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1: Input: composite  $N$ 
2: Pick  $a \in \{2, \dots, N-1\}$  uniformly at random
3: if  $g \leftarrow \gcd(a, N) > 1$  then
4:   return  $(g, N/g)$ 
5: end if
6: [Quantum] run QPE on  $U_a$  with  $t \geq 2\lceil \log_2 N \rceil + 1$ 
   counting qubits and read measurement  $m_\phi$ 
7: Use continued fractions to extract  $r$  from  $m_\phi/2^t$ 
8: if  $r$  is odd or  $a^{r/2} \equiv -1 \pmod{N}$  then
9:   restart at line 2 with a fresh  $a$ 
10: end if
11: return  $(\gcd(a^{r/2} - 1, N), \gcd(a^{r/2} + 1, N))$ 

```

multiplication construction reports circuit depth $\mathcal{O}(n^3)$ and gate count $\mathcal{O}(n^3 \log n)$ [28]. Putting the comparison in stark contrast with the classical asymptotics of Sections III and III-C:

The qualitative point is that quantum factoring is *polynomial-time*; classical factoring is *sub-exponential at best*. This is the reason RSA is considered vulnerable to a sufficiently large, fault-tolerant quantum computer.

V. EXPERIMENTAL DEMONSTRATION ON IBM HERON R2

A. Setup

We instantiate Shor's algorithm for $N = 15$, $a = 7$, with $n_{\text{count}} = 4$ counting qubits and 4 work qubits (since $15 < 2^4$). The modular multiplications $7^{2^k} \bmod 15$ are compiled as small permutations of the work register following the standard small- N Qiskit exposition [29]. The complete circuit has 8 qubits and a textbook depth of 7 before transpilation. The base $a = 7$ has order $r = 4$ modulo 15.

B. Circuit Structure

Figure 3 shows the top-level structure of the compiled Shor circuit: Hadamards on the counting register, an $|1\rangle$ initialisation of the work register, controlled- $U_a^{2^k}$ operations, and an inverse QFT on the counting register followed by measurement. Figure 4 opens the two sub-circuits — the inverse QFT and the modular multiplier — that constitute the non-trivial parts of the construction. Figure 5 shows the bare modular-multiplication-by-7 permutation on the work register.

C. Test Case A: Local Noiseless Simulation

We first establish ground truth by executing the circuit on the `qiskit-aer` statevector-derived sampler with 4096 shots and the simulator's native gate set. The measurement distribution (Fig. 6) places essentially all mass on the four bitstrings $\{0000, 0100, 1000, 1100\}$, corresponding to phase integers $m_\phi \in \{0, 4, 8, 12\}$ and hence phases $s/4$. Continued-fraction post-processing of the $m_\phi = 4$ or 12 peaks gives the full order $r = 4$; the $m_\phi = 8$ peak reduces to denominator 2 and, in this small instance, still yields a non-trivial factor during the classical post-processing. Using the full order, the usual reduction produces

$$\gcd(7^2 - 1, 15) = \gcd(48, 15) = 3, \quad (12)$$

$$\gcd(7^2 + 1, 15) = \gcd(50, 15) = 5. \quad (13)$$

The circuit construction is therefore mathematically correct.

D. Test Case B: Real IBM Heron r2 Hardware

We then execute the *same logical circuit* on `ibm_kingston`, a 156Q IBM Heron r2 backend announced by IBM in 2025 [6], via Qiskit Runtime, transpiling with `optimization_level=3` for the native basis $\{\text{sx}, \text{rz}, \text{cz}, x, \text{measure}\}$ and the chip's heavy-hex connectivity. The transpiled circuit grows to 156 qubits with a depth of approximately 4,021 and roughly 1,463 two-qubit `cz` gates — already deep enough that gate, readout, and decoherence errors substantially redistribute the output mass. Sampling with 4096 shots produces the histogram in Fig. 7. In this run, the bin corresponding to $m_\phi = 8$ (binary 1000) appears among the high-frequency outcomes; continued-fraction post-processing maps it to denominator 2, which is not the full order of 7 modulo 15 but is still sufficient in this toy instance to recover a non-trivial factor. The post-processor returns the correct factorisation $15 = 3 \times 5$. Reconstructing $\varphi(N) = 8$ then yields $d = 3$, which decrypts the ciphertext to the original plaintext $m = 8$. Figure 8 overlays the simulator and hardware distributions.

E. Historical Context

Experimental implementations of Shor's algorithm at small moduli have a 25-year history: Vandersypen *et al.* [30] factored 15 in 2001 using liquid-state nuclear magnetic resonance; Politi *et al.* [31] did so in 2009 on a photonic chip; Martín-López *et al.* [32] demonstrated

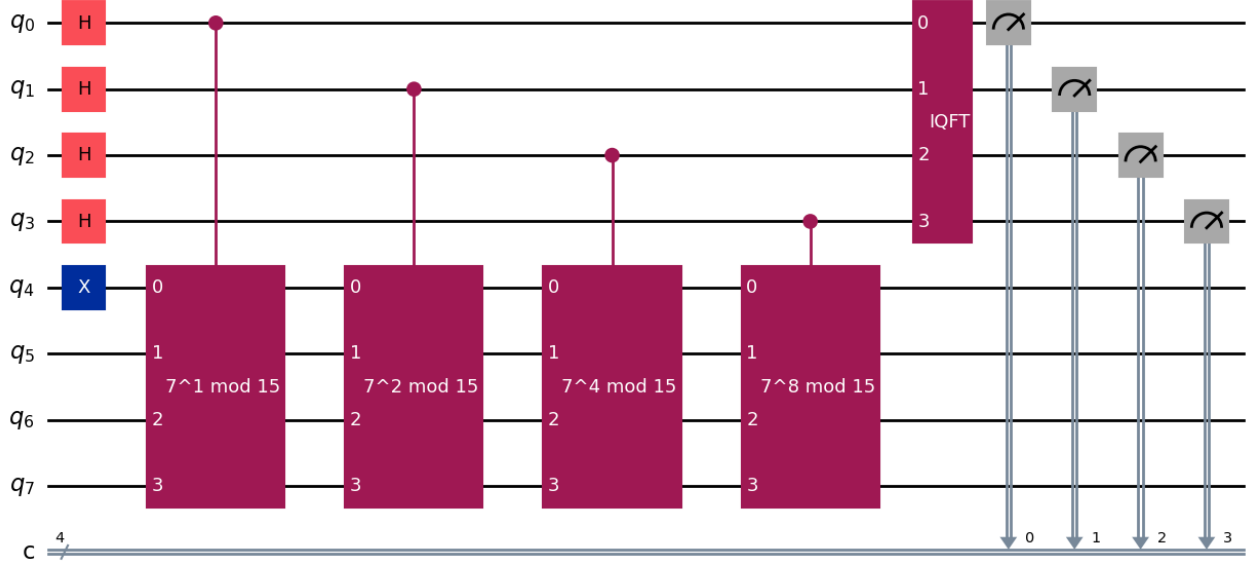


Fig. 3: Top-level Shor circuit for $N = 15$, $a = 7$, $n_{\text{count}} = 4$. Blocks labelled $7^{2^k} \bmod 15$ are controlled modular-multiplication unitaries; the final block on the counting register is the inverse Quantum Fourier Transform.

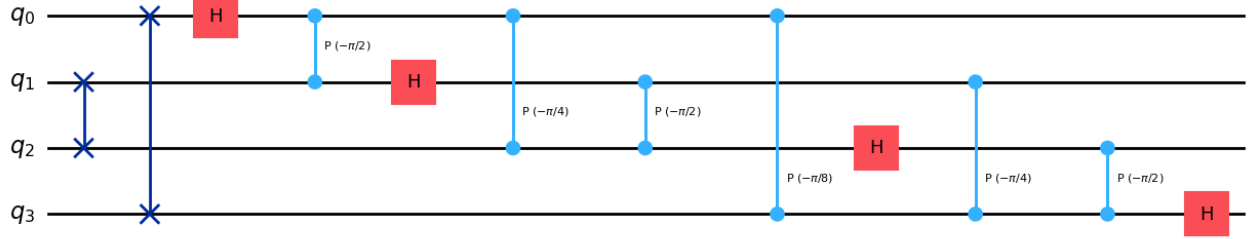


Fig. 4: The inverse-QFT and modular-multiplication sub-circuits viewed separately. The inverse QFT performs the phase estimation; the modular multiplier is the periodic function whose period is being measured.

qubit recycling for $N = 21$ in 2012; Lucero *et al.* [33] factored 15 with a Josephson phase qubit in the same year. The present demonstration is in that lineage: it does not factor a larger modulus, and that limitation is the central engineering story we now discuss.

VI. WHY $N = 15$ AND NOT RSA-2048?

The qubit budget of contemporary devices is, in isolation, no longer the binding constraint for small-modulus demonstrations: a 156-qubit Heron r2 chip has enough physical qubits to host small demonstrations and some space-optimised toy circuits. The limiting issue is circuit depth and error accumulation. Space-optimised constructions still require polynomial-depth modular arithmetic, and the Häner-Roetteler-Svore design reports $\mathcal{O}(n^3)$ depth and $\mathcal{O}(n^3 \log n)$ gate count [28]. For RSA-2048, published and preprint resource estimates require fault tolerance rather than direct NISQ execution [8], [9].

The path to a cryptographically relevant attack therefore requires quantum error correction. The leading near-term proposal is the *surface code* [34], whose physical-qubit overhead per logical qubit grows polynomially with the required logical error rate. One influential estimate for RSA-2048 is that of Gidney and Ekerä [8]: roughly 2×10^7 noisy physical qubits arranged into a surface-code factory, executing for approximately eight hours. A later Gidney preprint reduces the qubit estimate to below one million noisy physical qubits, while extending the runtime to below one week under similar assumptions [9]. Either estimate is far beyond a 156-qubit NISQ processor and assumes fault-tolerant operation. This gap explains why RSA is not broken by current NISQ devices.

VII. THE QUANTUM-CRYPTOGRAPHIC RISK TIMELINE

A. Harvest-Now, Decrypt-Later

The asymmetry between the construction of a cryptographically relevant quantum computer (CRQC) and the

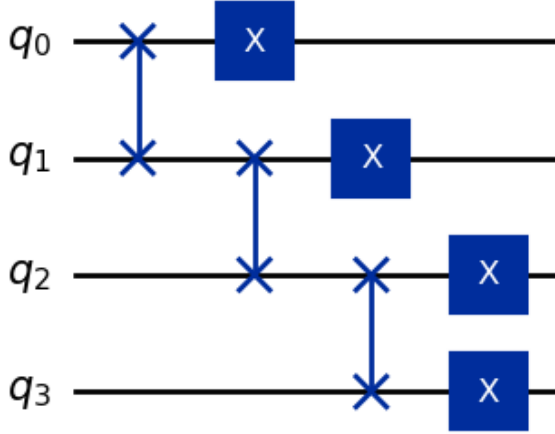


Fig. 5: The bare permutation implementing $|y\rangle \mapsto |7y \bmod 15\rangle$ on the four work qubits, consisting of three SWAPs and four Pauli- X gates.

half-life of the data that today’s cryptography protects gives rise to the *harvest-now, decrypt-later* (HNDL) threat. An adversary need not possess a CRQC *today* in order to threaten today’s encrypted traffic; they need only *archive the ciphertext now* and decrypt it once a CRQC becomes available. For data whose confidentiality requirement extends decades into the future — intelligence, medical, diplomatic, intellectual-property, and identity-related material — the relevant cryptographic horizon is the *sum* of (a) the time during which the data must remain confidential and (b) the time required for an adversary to migrate to post-quantum protections. Mosca [10] formalised this as the inequality

$$X + Y > Z, \quad (14)$$

where X is the required confidentiality duration, Y is the migration time to post-quantum cryptography, and Z is the time until a CRQC exists. If $X + Y > Z$, then *the cryptography is already broken from the adversary’s perspective*.

B. Resource Estimates and the Engineering Frontier

One detailed resource-cost estimate for RSA-2048 is the 2021 Gidney–Ekerä result: ~ 8 hours of wall-clock execution on $\sim 2 \times 10^7$ noisy physical qubits at a physical gate-error rate near 10^{-3} [8]. A 2025 preprint by Gidney reports a lower qubit count — less than one million noisy physical qubits — with a longer runtime of less than one week under similar assumptions [9]. Several caveats apply: these are model-dependent estimates involving particular surface-code layouts, magic-state or related resource factories, and noise assumptions. They should be read as resource estimates, not deadlines.

C. Post-Quantum Cryptographic Standards

The defensive response is migration to schemes whose hardness assumptions are not known to fall to a quantum adversary. The U.S. National Institute of Standards and Technology published three post-quantum cryptography standards in August 2024:

- **FIPS 203** [11], Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM, based on CRYSTALS-Kyber);
- **FIPS 204** [12], Module-Lattice-Based Digital Signature Standard (ML-DSA, based on CRYSTALS-Dilithium);
- **FIPS 205** [13], Stateless Hash-Based Digital Signature Standard (SLH-DSA, based on SPHINCS⁺).

The U.S. National Security Agency’s Commercial National Security Algorithm Suite 2.0 [17] publishes migration guidance for National Security Systems, while National Security Memorandum NSM-10 [16] directs federal agencies to inventory cryptographically dependent systems and prepare for migration. Bennett and Brassard’s BB84 quantum key distribution protocol [35] remains a hardware-rooted path to information-theoretic key agreement, although it is not a drop-in replacement for all public-key infrastructure.

VIII. PUBLIC POLICY AND RESEARCH CONTEXT

Public quantum-computing programmes are now tied directly to cryptographic migration planning. In the United States, the National Quantum Initiative Act created a coordinated federal quantum research programme [36], while NSM-10 directed agencies to account for systems that depend on cryptography vulnerable to a future CRQC [16]. NSA’s CNSA 2.0 guidance provides a migration framework for National Security Systems [17]. In Europe, the Quantum Technologies Flagship coordinates quantum research across computing, communication, simulation, and sensing [37].

Scientific demonstrations also show rapid hardware progress, but they should not be confused with RSA-breaking capability. Google’s Sycamore experiment reported a sampling-based quantum advantage result in 2019 [38]. The Jiuzhang photonic experiment and the Zuchongzhi superconducting processor reported different quantum-advantage benchmarks in 2020 and 2021, respectively [39], [40]. These experiments are important for quantum information science, but they do not implement a fault-tolerant RSA-2048 attack.

IX. A DOUBLE-EDGED SWORD

Quantum computation is a clear dual-use technology. On the defensive and constructive side, fault-tolerant quantum processors may improve some simulations in quantum chemistry, materials, and condensed-matter physics [41]. Quadratic speed-ups via Grover’s algorithm [42] apply broadly to search and combinatorial problems. Quantum key distribution in the BB84

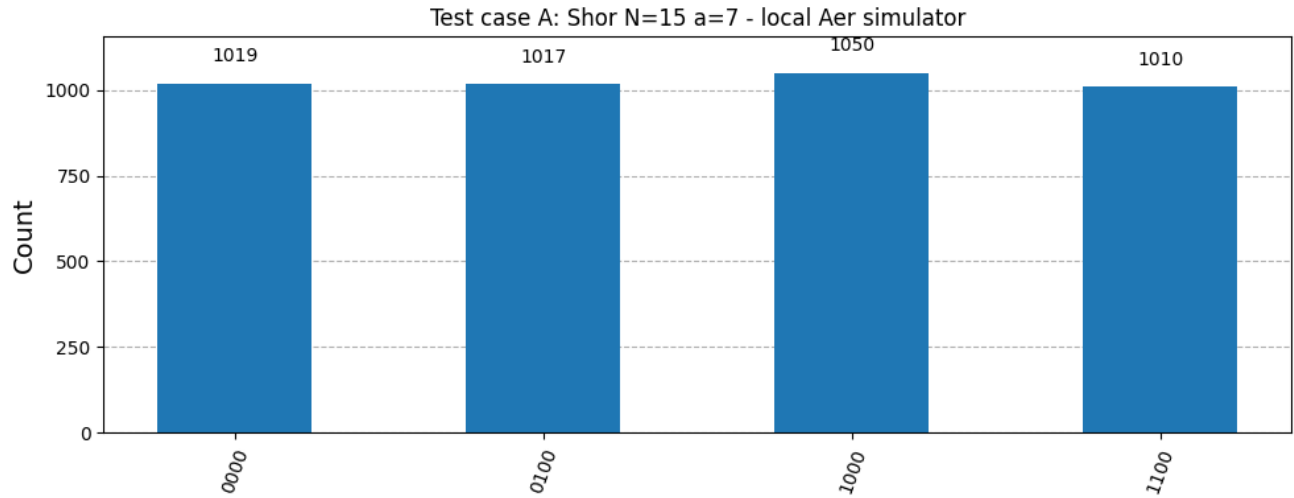


Fig. 6: Test case A. Measurement histogram from the noiseless local Aer simulator (4096 shots) for $N = 15$, $a = 7$. Mass concentrates on the four canonical phase integers $\{0, 4, 8, 12\}$; reduced-fraction post-processing of these peaks supplies either the full order or, for this toy instance, a usable divisor that still reveals a factor.

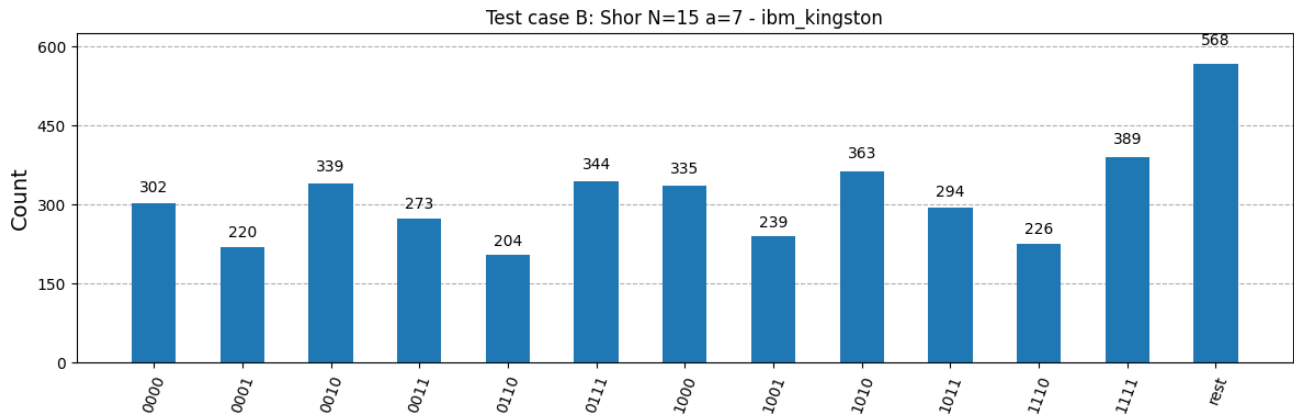


Fig. 7: Test case B. Measurement histogram from `ibm_kingston`, a 156-qubit IBM Heron r2 superconducting processor (4096 shots). Hardware noise broadens the distribution substantially; in this run, the 1000 bin remains frequent enough for the classical post-processor to recover a non-trivial factor.

tradition [35] provides information-theoretic, rather than computational, security guarantees.

On the offensive side, the same machine that runs a chemistry simulation can run Shor’s algorithm. The security premise of RSA, finite-field Diffie–Hellman, and elliptic-curve cryptography would fail once a sufficiently large fault-tolerant quantum computer can run Shor’s algorithm [5]. The double-edged character is not avoidable; it is structural. The correct posture for institutions, governments, and operators is neither prohibition (which would forfeit the constructive gains) nor complacency (which would forfeit the defensive runway), but *timely cryptographic migration*, instrumented by Mosca’s inequality and guided by the NIST post-quantum standards [11]–[13].

X. SCOPE AND LIMITATIONS OF THIS ARTICLE

To be explicit about what this work does and does not claim:

- **No novel cryptographic result is claimed.** Every algorithm, theorem, and resource estimate cited here exists in the prior literature. The contribution is pedagogical and demonstrative.
- **The experiment is at $N = 15$, not at any cryptographically meaningful modulus.** As Sections VI and VII make explicit, the gap to RSA-2048 is many orders of magnitude in fault-tolerant resources, as quantified by Gidney and Ekerä [8].
- **Boneh’s surveyed classical attacks on RSA** [43] are orthogonal to this article and not discussed in detail; the focus here is the asymptotic factoring story.
- **The experiment used a standard small- N modular-multiplication permutation** [29], not

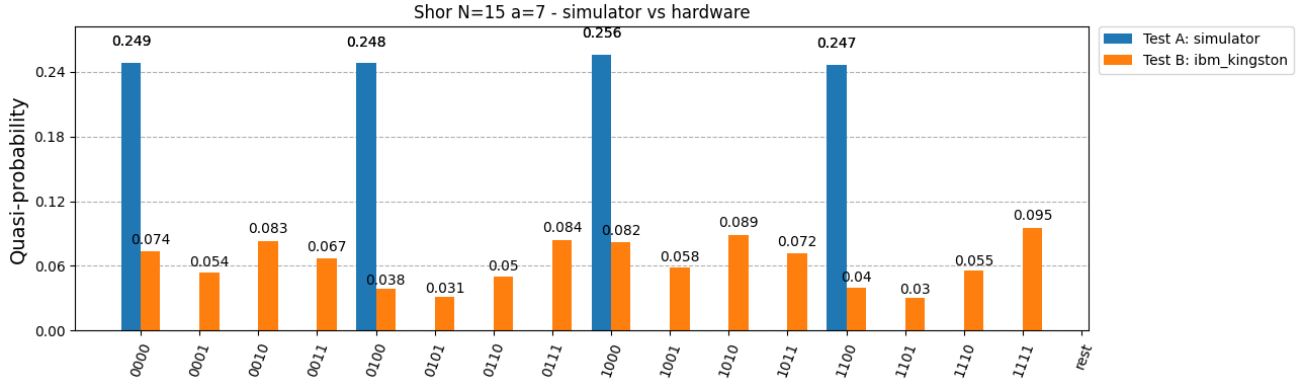


Fig. 8: Side-by-side normalised measurement distributions for test cases A (noiseless simulator) and B (Heron r2 hardware). The simulator concentrates mass on the four canonical phase integers; the hardware distribution is much broader, but the post-processing step still finds a usable factor-producing outcome in this recorded run.

a fully general modular multiplier; a 2048-bit attack would require the Beauregard [27] or Häner–Roetteler–Svore [28] constructions.

- **The reported local and hardware measurements are notebook-backed.** The RSA setup, classical factoring measurements, simulator counts, hardware job metadata, and hardware post-processing are documented in the accompanying notebooks referenced below. Results attributed to external researchers are cited to the original publications instead.

XI. CONCLUSION

We have walked through the cryptographic mathematics of RSA, the classical-cryptanalytic landscape that protects it today, and the quantum-cryptanalytic landscape that threatens it in the long term. We accompanied the theoretical treatment with a notebook-documented execution of Shor’s algorithm at $N = 15$ on a real 156-qubit IBM Heron r2 superconducting processor. In the recorded run, the post-processing recovered the prime factors $\{3, 5\}$ despite gate, readout, and decoherence noise. The demonstration is small, and intentionally so: it shows small-scale behaviour honestly, rather than overstating it. The right conclusion to draw is neither alarm nor dismissal, but *movement* — toward post-quantum standards already published, toward cryptographic inventories already mandated, and toward the long engineering programme required for fault-tolerant quantum computation.

CODE AND NOTEBOOK AVAILABILITY

The project repository for the notebooks and supporting files is https://github.com/mlesterdampios/shors_algorithm_for_breaking_rsa. The notebooks used for the RSA setup, classical factoring tests, quantum simulation, IBM hardware execution, generated figures, and post-processing are the sources for the experimental values reported here. Unless a result is explicitly attributed to another researcher or publication, experimental values in

this article are taken from those notebooks. No claim is made that the $N = 15$ demonstration constitutes a cryptographically meaningful RSA break.

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